Lake Yang

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Visiting Researcher at Imperial College London (Complex Multiscale Systems Group, Prof. Serafim Kalliadasis) and Quantitative Analyst at UBS, working on physics-informed and structure-preserving machine learning for dynamical systems and PDE simulation. First author of two papers: MPINeuralODE (ICML 2026 AI4Physics Workshop), on learning globally consistent dynamics with physics-informed neural ODEs, and GeomHerd (AAAI 2027, Phase 2 review), on geometric analysis of multi-agent systems. Both projects were carried out alongside a full-time role at UBS; applying for part-time doctoral study (2027 entry).

RESEARCH INTERESTS

My research centres on structure-preserving and physics-informed machine learning for high-dimensional dynamical systems: learning models that generalise beyond the trajectories they were trained on, respect conservation laws over long horizons, and remain stable under PDE-based simulation. I am also interested in the geometry of multi-agent interaction, where curvature and Ricci flow on interaction graphs give early-warning signals of collective behaviour such as herding and systemic risk. Methods: neural ODEs, physics-informed neural networks, geometric and graph-based learning, multi-agent simulation; applications in fluid dynamics, electromagnetics and complex systems.

RESEARCH EXPERIENCE

Imperial College London **August 2025 – Present**

Visiting Researcher — Complex Multiscale Systems Group, Dept. of Chemical Engineering *London, United Kingdom*

- Supervised by Prof. Serafim Kalliadasis (Imperial profile).
- **MPINeuralODE** *First author · accepted, ICML 2026 AI4Physics Workshop*
  - Addresses the generalisation failure of neural ODEs, which overfit single trajectories rather than learning the underlying dynamics.
  - Proposed a multiple-initial-condition curriculum with a soft physics-residual collocation loss, evaluated jointly on out-of-sample accuracy, long-horizon stability and Hamiltonian conservation.
  - Designed the methodology, implemented the full codebase from scratch (public repository) and trained all models, using a compact 4×128 network ($\sim$67k parameters).
  - Co-authored with members of the Complex Multiscale Systems Group under Prof. Kalliadasis; prepared the accepted workshop paper and its camera-ready version, including the arXiv preprint.
- **GeomHerd** *First author · AAAI 2027, advanced to Phase 2 review*
  - Forward-looking herding detection that measures Ollivier–Ricci curvature and Ricci-flow dynamics on agent-interaction graphs from multi-agent simulations, linked to equilibrium selection.
  - Constructs interaction graphs from windowed action agreement, defines lazy random-walk transition kernels and a mean-curvature signal, and identifies a singularity time in the curvature dynamics; a mean-field proposition connects the geometric signal to the cross-sectional absolute deviation (CSAD) herding measure used in empirical finance.
  - Applications to algorithmic herding, crowding and AI-monoculture systemic risk. Preprint available on arXiv.

Johns Hopkins University — *Research Assistant (Prof. Fadil Santosa), remote* *April 2023 – April 2024*

- Designed and implemented a physics-informed neural network (PINN) from scratch to solve Maxwell's equations for EM-wave interaction with a conductive sheet; methodological precursor to MPINeuralODE.

Massachusetts Institute of Technology (MITx) **January 2023 – December 2023**

Teaching Assistant, 2.25.1x Advanced Fluid Mechanics *Remote*

PUBLICATIONS

Yang, L., Malpica-Morales, A., Papadakis-Wood, F.-I., Kalliadasis, S. (2026). MPINeuralODE: Multiple-Initial-Condition Physics-Informed Neural ODEs for Globally Consistent Dynamical System Learning. *Accepted at the AI4Physics Workshop, ICML 2026.* arXiv: 2605.13305; OpenReview.

Yang, L., Su, J., Zeng, J., Lu, W., Qian, X., Zhang, W., Wu, C., Jin, D. (2026). GeomHerd: A Forward-looking Herding Quantification via Ricci Flow Geometry on Agent Interactive Simulations. *Under review at AAAI 2027 (advanced to Phase 2).* arXiv: 2605.11645.

Fan, Y., Jin, C., **Yang, J.**, Zhu, F., Li, W. (2024). Single-metalens-assisted polarization imaging and edge detection for target recognition. *Appl. Phys. Lett.* 124, 181702. DOI: 10.1063/5.0192217. Carried out with the Changchun Institute of Optics, Fine Mechanics and Physics (CIOMP), Chinese Academy of Sciences.

HONOURS

- TechWomen100 Award Winner 2026, recognising the UK's leading women in technology.
- Winner, UBS Asset Management AI Hackathon (2025).
- First junior at UBS to achieve Gold Certified Engineer, Certified Data Scientist and Certified Data Analyst (2024).
- University of Manchester: Department of Mathematics Scholarship (2021); Volunteer of the Year, Highly Commended, four consecutive years (2021–2024).

EDUCATION

University of Manchester **September 2020 – June 2023**

BSc (Hons) Mathematics with Financial Mathematics — First Class, GPA 3.92/4.0 *Manchester, United Kingdom*

- Core modules: PDEs, Stochastic DEs, Green's Functions, Fluid Mechanics, Probability, Statistics, Multivariate Statistics & Machine Learning, Linear Algebra, Financial Modelling.
- Master's offers: Part III MASt Applied Mathematics (Cambridge), Applied Mathematics (Columbia), Applied Mathematics (Imperial College London).
- Further study: IBM AI Engineering Professional Certificate; IBM RAG & Agentic AI Professional Certificate; Financial Services Professional Level 6 Apprenticeship, Fitch Learning (in progress); CFA Level I candidate.

PROFESSIONAL EXPERIENCE

UBS AG **June 2022 – Present**

Quantitative Analyst *London, United Kingdom*

- **Quantitative Analyst**, *Investment Banking — Global Markets* September 2025 – Present
  - Sole owner and developer of the production Prime Brokerage AI platform: an agentic system on LlamaIndex with a reusable MCP analytical harness over governed data, live for sales, coverage and financing users across EMEA, Americas and APAC.
  - Primary developer of the Capital Introduction AI platform: a Graph-RAG multi-agent system (LangGraph, Neo4j) with hybrid retrieval over call notes, fund documents and structured data, returning citation-traceable answers.
  - Designed a human-in-the-loop AutoResearch framework: an LLM constructs calculation logic from a natural-language data requirement, the business validates it, and approved logic persists as a reusable endpoint.
  - Built semantic, entity-resolution and tool-routing layers mapping natural-language questions to governed NL2SQL, causality and driver-decomposition workflows, validated against existing dashboards and desk feedback.
- **Quantitative Analyst**, *Asset Management — Investing* February 2025 – September 2025
  - Built the primary RAG pipeline over S&P 500 research corpora (hierarchical chunking, BM25 + dense retrieval via pgvector, multi-intent router); won the UBS Asset Management AI Hackathon.
  - Developed an ML/statistical engine blending multiple ESG sources into a single Controversy Score, reducing source redundancy and licensing cost while improving alignment with historical price momentum.
- **Quantitative Analyst**, *Global Markets — Rates, then Cash Equities* September 2024 – February 2025
  - Built rates analytics tooling on the Fixed Income & Rates desk, then led cross-asset AI initiatives in Cash Equities.
  - Built a metadata-driven Text-to-SQL model with schema-aware prompting and automated query repair, lifting SQL-generation accuracy from 76% to 93%.
  - Designed a three-stage SEC 10-K ESG NLP cascade (FinBERT-ESG → ClimateBERT-F → FinBERT sentiment/greenwashing); fine-tuned `bert-base-uncased` on a UBS-curated corpus, lifting accuracy from 45% to 98.3% and expanding ESG coverage 5.6×.
- **Software Engineer**, *Investment Banking — Market Regulation* September 2023 – September 2024
  - Led the migration of 3B+ trading records from Oracle to Delta Lake via Spark ETL with custom reconciliation, delivered in five months; implemented column-level data lineage (∼80% deal-store coverage); Redis caching cut cloud query and storage cost by ∼60%.
  - *Summer Technology Analyst (June – September 2022):* built a real-time Power BI application that became the original version of the Certified Engineer Program, UBS's largest internal learning platform.

TECHNICAL SKILLS

ML & Scientific ML: neural ODEs (torchdiffeq), physics-informed neural networks (DeepXDE), Hamiltonian-conserving models; Ricci-flow graph geometry (Ollivier–Ricci curvature); agent-based simulation and herding measures (CSAD); reinforcement learning (PPO, SAC, multi-agent RL); CNNs (ResNet, VGG); gradient boosting (LightGBM, XGBoost), random forests.

Scientific Computing: PDE simulation (fluid dynamics, Maxwell's equations), stochastic DEs, Monte Carlo methods; experimental rigour for time-series models (purged walk-forward cross-validation, Deflated Sharpe ratio, probability of backtest overfitting).

Programming & Engineering: Python, SQL, Java, R, MATLAB, Bash, LaTeX; PyTorch, TensorFlow; FastAPI, Pydantic; Spark, Databricks, Delta Lake; AWS, Azure; Docker, Kubernetes, MLflow, Git, CI/CD.

LLM & Agentic Systems: MCP servers, LangGraph, LangChain, LlamaIndex; Graph RAG (Neo4j), hybrid retrieval; NL2SQL; fine-tuning (Hugging Face Transformers, FinBERT).

Languages: English (fluent), Mandarin (native), German (intermediate).

LEADERSHIP & COMMUNITY

Lead, Educate Stream, Women in Tech, UBS: designed and delivered a firm-wide AI training programme (agentic AI, RAG, LLM systems) and host internal tech conferences broadening access to AI and quant careers. Earlier, presented my summer-placement Power BI application at the London Data Community.

Founder, The Blue Letters Project (November 2021 – Present): lead the charity across UK universities, managing all volunteer managers and 700+ volunteers on record delivering mental-health support and community outreach.